

Spectrometry Lab: Determining Thermodynamic Properties

Patryk Kozlowski
patryk_kozlowski@g.harvard.edu

Lab 4 - General Chemistry

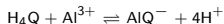
Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:

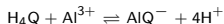


- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:

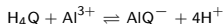


- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:

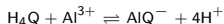


- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



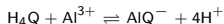
- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Key Calculations:

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

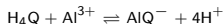
Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

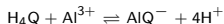
Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

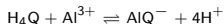
Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

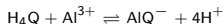
Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$
- ▶ Use $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ to determine ΔH° and ΔS° from a plot of ΔG° vs. T .

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$
- ▶ Use $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ to determine ΔH° and ΔS° from a plot of ΔG° vs. T .

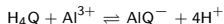
Experiment Steps:

- ▶ Measure initial absorbance A_i at 550 nm.

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$
- ▶ Use $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ to determine ΔH° and ΔS° from a plot of ΔG° vs. T .

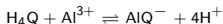
Experiment Steps:

- ▶ Measure initial absorbance A_i at 550 nm.
- ▶ Heat the sample and record absorbance at different temperatures (90°C to 50°C).

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$
- ▶ Use $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ to determine ΔH° and ΔS° from a plot of ΔG° vs. T .

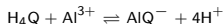
Experiment Steps:

- ▶ Measure initial absorbance A_i at 550 nm.
- ▶ Heat the sample and record absorbance at different temperatures (90°C to 50°C).
- ▶ Use absorbance data to find $[\text{AlQ}^-]$ and calculate K at each temperature.

Lab 4: Spectrophotometry and Thermodynamics

Objective: Use spectrophotometry to measure equilibrium concentrations at different temperatures and determine thermodynamic properties like ΔG° , ΔH° , and ΔS° .

Reaction: Complexation of Aluminum Ions with Xylenol Orange:



- ▶ Yellow reactant (H_4Q) forms red product (AlQ^-).
- ▶ Measure absorbance at 550 nm (product absorbs more strongly at this wavelength).
- ▶ The reaction is endothermic (heat is required to drive the reaction forward).

Key Calculations:

- ▶ Beer's Law: $A = \epsilon_1[\text{H}_4\text{Q}]L + \epsilon_2[\text{AlQ}^-]L$
- ▶ Find equilibrium constant K at each temperature.
- ▶ Calculate ΔG° from K : $\Delta G^\circ = -RT \ln K$
- ▶ Use $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ to determine ΔH° and ΔS° from a plot of ΔG° vs. T .

Experiment Steps:

- ▶ Measure initial absorbance A_i at 550 nm.
- ▶ Heat the sample and record absorbance at different temperatures (90°C to 50°C).
- ▶ Use absorbance data to find $[\text{AlQ}^-]$ and calculate K at each temperature.